Lab Project:
Secure File transfer using protocol SFTP.

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Secure File Transfer Using SFTP

Objective:

To securely transfer files between a Windows Server and client using SFTP (SSH File Transfer Protocol).

Prerequisites:

- A Windows machine with WinSCP installed.
- A Linux machine with an SSH server running.
- Network connectivity between the Windows and Linux machines.

Lab Setup:

1.Windows Server:

- Install WinSCP on windows from the following link.

https://winscp.net/eng/download.php

2.Linux Server:

- Install OpenSSH Server on Linux (If Not Installed)

sudo apt update && sudo apt install openssh-server -y	For Debian/Ubuntu
sudo systemctl enable ssh && sudo systemctl start ssh	Enable and start SSH

- Verify SSH is running:

```
systemctl status ssh
```

```
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform: ~
ervice.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/l
ib/systemd/system/ssh.service.
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ sudo systemctl start ssh
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab>
     Active: active (running) since Tue 2026-03-10 12:23:49 PKT; 6min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 4929 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 4930 (sshd)
      Tasks: 1 (limit: 2220)
     Memory: 1.2M (peak: 1.5M)
        CPU: 140ms
     CGroup: /system.slice/ssh.service
             └─4930 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform systemd[1]: Starting ssh.>
Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform sshd[4930]: Server listen>
Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform sshd[4930]: Server listen>
Mar 10 12:23:49 hadiya-zuberi-VMware-Virtual-Platform systemd[1]: Started ssh.s>
lines 1-18/18 (END)
```

- Find the Linux machine's IP address:

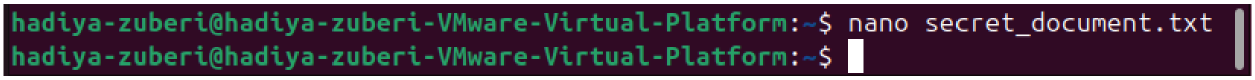
ip a | grep inet or ifconfig

```
hadiya-zuberi@hadiya-zuberi-VMware-Virtual-Platform:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP g
roup default qlen 1000
    link/ether 00:0c:29:29:39:c6 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.205.131/24 brd 192.168.205.255 scope global dynamic noprefixrou
te ens33
       valid_lft 973sec preferred_lft 973sec
    inet6 fe80::20c:29ff:fe29:39c6/64 scope link
       valid_lft forever preferred_lft forever
```

Task:

(Before Proceeding create a text file in linux add some data in it or download anything from internet so that you can transfer it to the windows machine consider that file a top secret level document)

nano secret_document.txt



Add some content, e.g.:

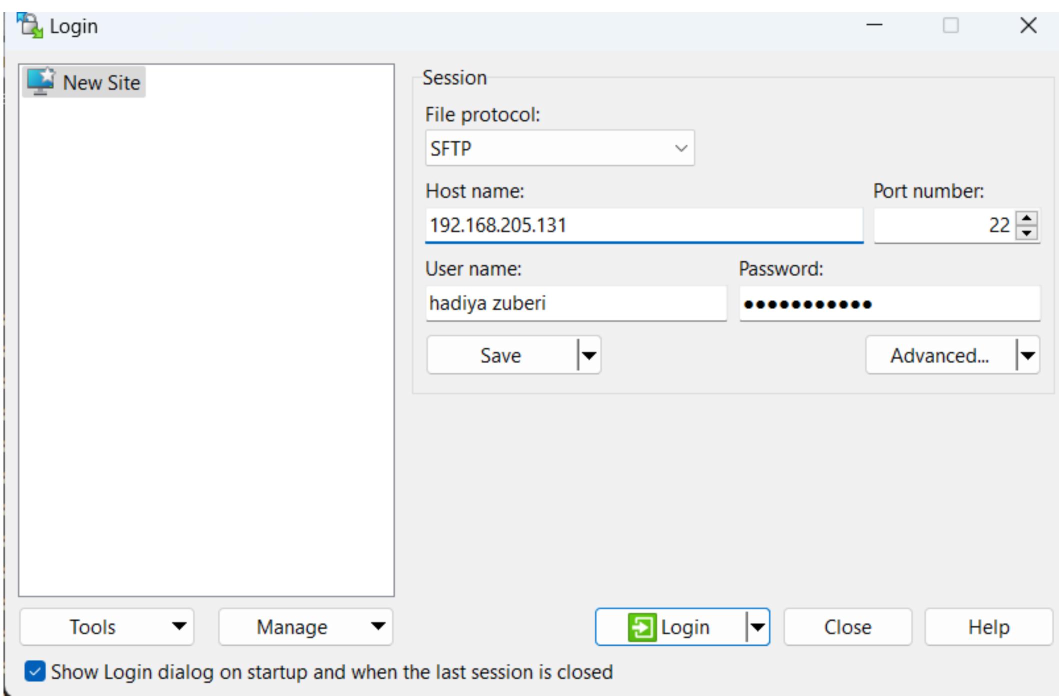
TOP SECRET - Lab Document

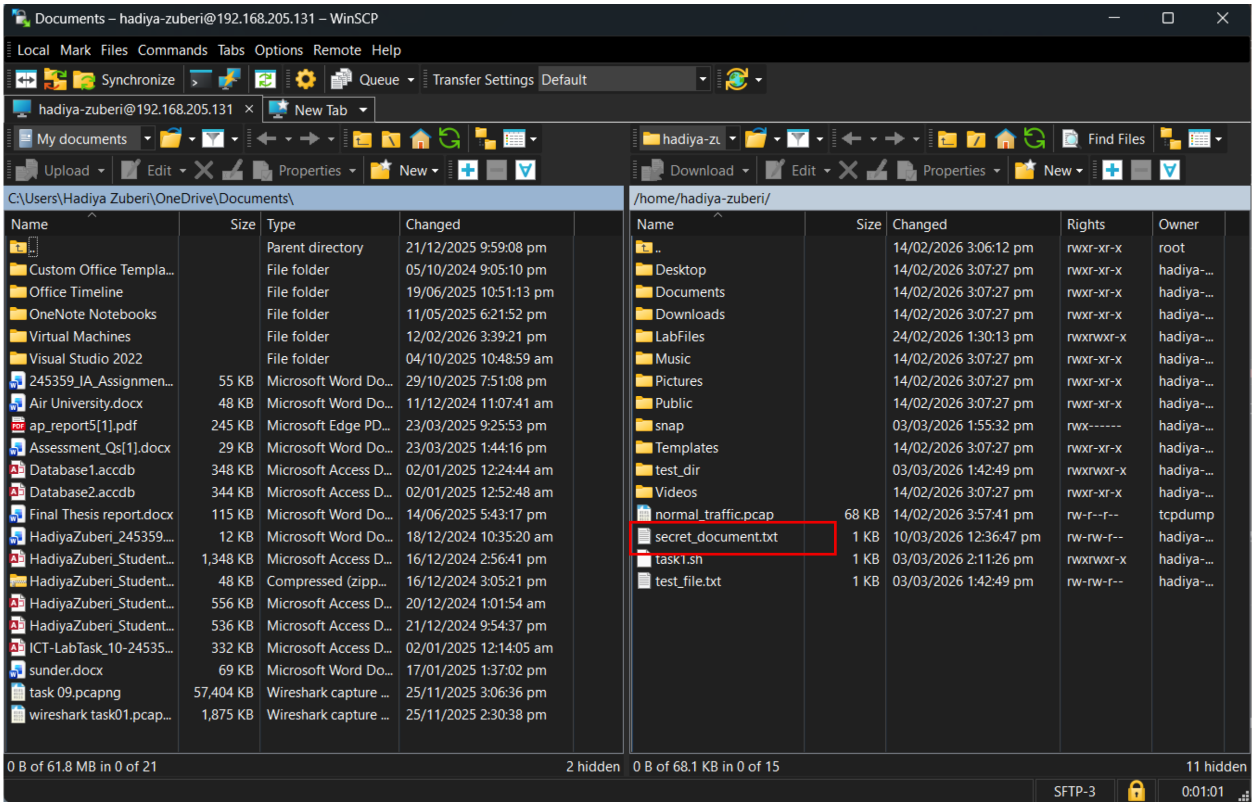
Project Code: ALPHA-7

Classified Data: 1234-ABCD

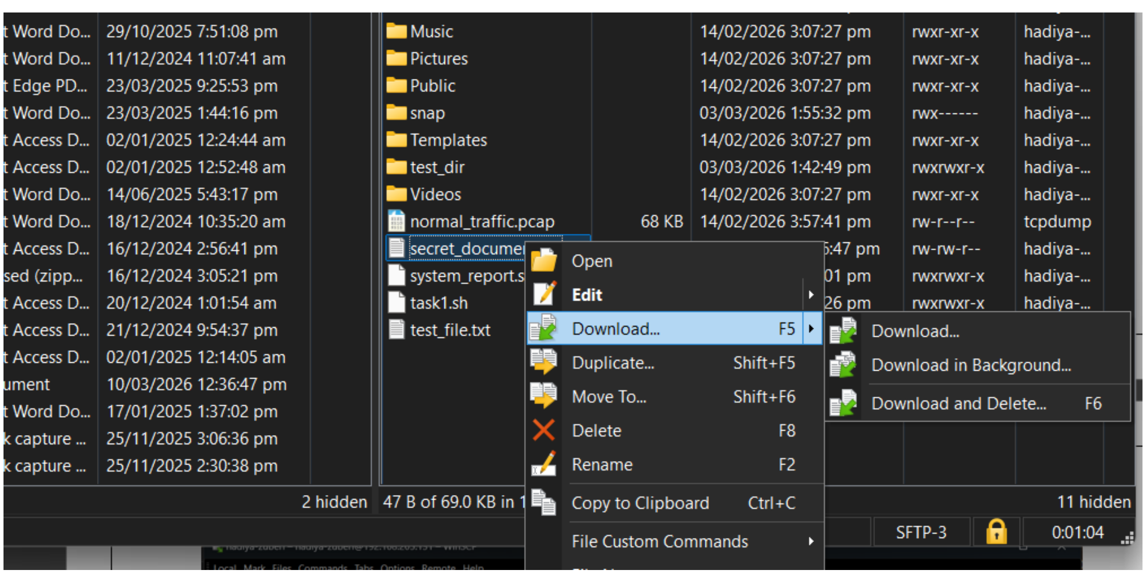
After adding the content save it with Ctrl + S and then exit using Ctrl + X

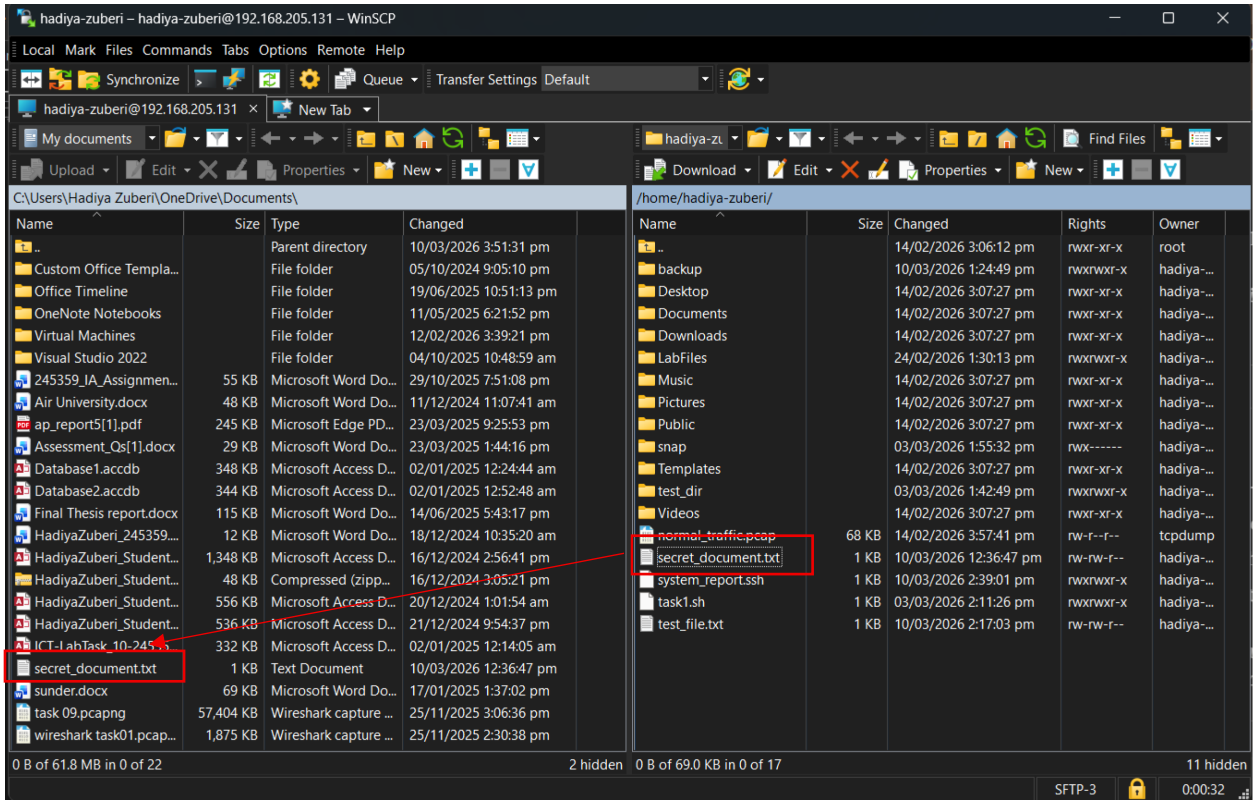
- ➢ Connect to Linux from Windows Using WinSCP
- Open WinSCP.
- Select SFTP as the file protocol.
- Enter the Linux machine's IP address in the Host Name field.
- Enter your Linux username and password.
- Click Login to connect.





- ➢ Download Files from Linux to Windows
- Once connected, navigate to the file location on the Linux machine in the remote panel.
- Navigate to the desired location on Windows in the local panel.
- Drag and drop files from Linux to Windows to download them. OR right click the desired file and click download.





Note:

- Ensure the file appears in the specified Windows folder.
- Open the file to confirm it transferred correctly.

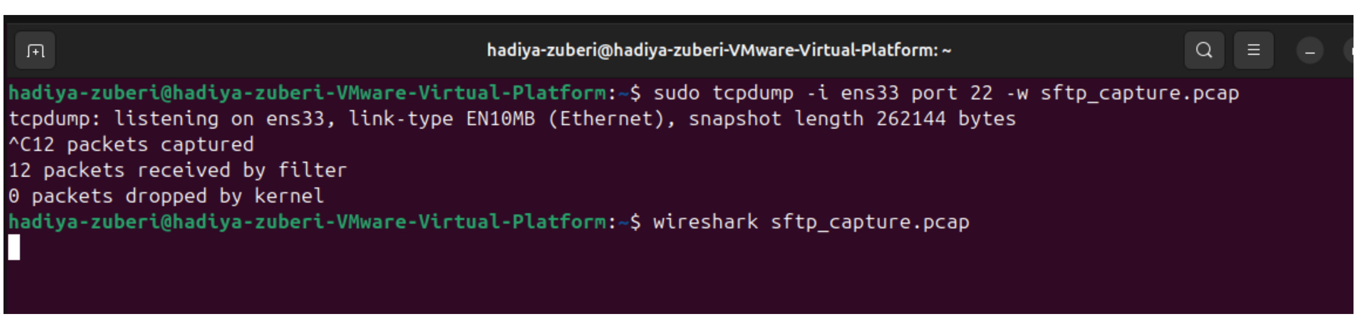
To prove the transfer was secure (encrypted):

1. Before transferring the file type this command in Ubuntu terminal to capture packets

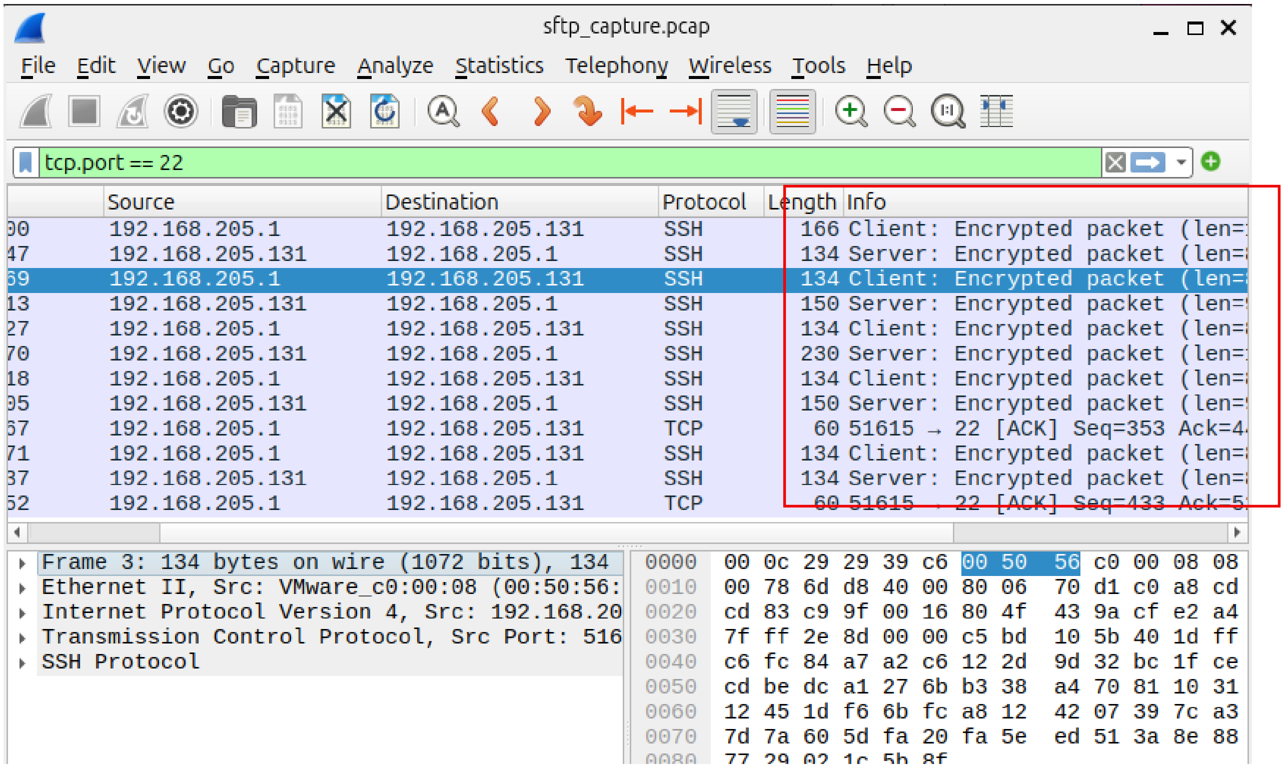
   sudo tcpdump -i ens33 port 22 -w sftp_capture.pcap

2. Transfer the file from WinSCP on windows. Go back to Ubuntu and press Ctrl + C to stop capture.
3. Open the pcap file in Wireshark

   Wireshark sftp_capture.pcap



4. Filter port 22 to check whether the packets were encrypted or not.



Conclusion

This lab demonstrated key network security technique. A classified document was securely transferred from Ubuntu to Windows using SFTP over WinSCP, with Wireshark confirming the data was fully encrypted on port 22 and unreadable in transit. Both protocols together provide strong protection against unauthorized access and data interception.